Milk Hill 2001 - Fibonacci, Golden Ratio and Spiral Tests

Full model equations, per-arm results, and failed tests are retained.

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1. Golden-ratio reference

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phi = (1 + sqrt(5))/2 = 1.618033988749895
For a golden logarithmic spiral with a quarter-turn growth factor phi: b_phi = 2 ln(phi) / pi =
0.306348962530.
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2. Spiral models

Archimedean: r(theta) = a + b theta. Logarithmic: r(theta) = a exp(b theta), fitted by ln(r) = ln(a) + b theta. Each arm was tested separately using its completed circle centers.

Arm	Arch R2	Arch RMSE ft	Log b	Log R2	Log RMSE ft	b-b_phi
0	0.000916	79.954	0.023179	0.000142	83.166	-0.283170
1	0.002945	85.974	-0.075657	0.001236	89.729	-0.382006
2	0.014307	91.041	0.156424	0.006392	94.495	-0.149925
3	0.000786	86.777	0.021577	0.000208	89.218	-0.284772
4	0.001989	82.846	-0.004199	0.000004	86.319	-0.310548
5	0.034243	76.433	-0.067891	0.020098	78.688	-0.374240

All per-arm R2 values are near zero for these simple radius-versus-angle models. The fitted logarithmic b values range from about -0.0757 to 0.1564, not around the golden reference 0.306349.

3. Diameter progression and Fibonacci recurrence

For diameters d_n ordered by slot within an arm, the direct Fibonacci recurrence test is epsilon_n = d_n - d_(n-1) - d_(n-2). The normalized residual is NRMSE_F = sqrt(sum epsilon_n^2 / sum(d_n - mean(d))^2). A good Fibonacci recurrence would require a small NRMSE; values substantially above 1 indicate residual error comparable to or larger than the sequence variance.

Arm	Linear R2	Exp R2	Power R2	Fib NRMSE	Mean |d_n/d_n-1 - phi|
0	0.01836	0.04062	0.06363	1.9453	1.3168
1	0.00232	0.00608	0.01093	2.2759	1.5495
2	0.00647	0.01610	0.01456	2.0969	1.5054
3	0.04537	0.02695	0.02428	1.9647	1.2754
4	0.00739	0.00498	0.00093	1.9183	2.0563
5	0.00090	0.00063	0.00027	1.9307	1.3193

Result: the completed diameter sequences do not satisfy a Fibonacci recurrence under this direct test. NRMSE_F ranges from approximately 1.918 to 2.276 across the six arms.

4. Sixfold structure

The arithmetic structure is exact: 1 + 6 x 68 = 409. Comparing same-slot circles after 60-degree arm alignment gives aggregate RMS radial SD 17.372 ft, angular SD 15.329 degrees, and diameter SD 12.386 ft. Thus the count is exactly sixfold while individual same-slot geometry is not a rigidly repeated copy.

Controlling mathematical finding

The completed geometry supports the exact six-arm population count and scale controls. Under the explicit model tests above, it does not support a golden logarithmic spiral coefficient or a Fibonacci diameter recurrence.